

Assignment – Week 5

Introduction to Machine Learning-IIT KGP Topics:

Logistic Regression, SVM, Kernel Functions, Kernel SVM TYPE

OF QUESTION: MCQ/MSQ

Number of Questions: 10

Total Marks: $10 \times 2 = 20$

Question 1

In Logistic Regression, the predicted probability is obtained by applying which function to the linear combination of features?

- A. ReLU Function
- B. Sigmoid Function
- C. Tanh Function
- D. Identity Function

Correct Answer: B

Detailed Solution: Logistic Regression uses the sigmoid function.

Question 2

A Logistic Regression model is underfitting the training data. Which of the following actions may help?

1. Add polynomial features
2. Increase model complexity
3. Remove useful features

- A. 1 only
- B. 1 and 2 only
- C. 2 and 3 only
- D. 1, 2 and 3

Correct Answer: B

Detailed Solution: Adding polynomial features and increasing complexity can reduce underfitting

Question 3

Suppose a logistic regression model predicts a probability of 0.82 for the positive class. Using a threshold of 0.5, what is the predicted class?

- A. Positive
- B. Negative
- C. Cannot determine
- D. Both classes

Correct Answer: A

Detailed Solution: $0.82 > 0.5$, so classify as positive.

Question 4

Which of the following statements about Support Vectors is true?

- A. They are always the misclassified points
- B. They are the points closest to the decision boundary
- C. They are the points farthest from the decision boundary
- D. They are removed before training

Correct Answer: B

Detailed Solution: Support vectors are closest to the separating hyperplane.

Question 5

A binary classification problem contains 200 training samples. After training an SVM, only 12 samples become support vectors. Which statement is correct?

- A. All 200 samples are needed during prediction
- B. Only the support vectors are required to define the decision boundary
- C. The model is invalid because too few support vectors were selected
- D. SVM always selects half the training data as support vectors

Correct Answer: B

Detailed Solution: The decision boundary depends only on support vectors.

Question 6

For a 5-class classification problem, how many SVM classifiers must be trained using the One-vs-All approach?

- A. 4
- B. 5
- C. 10
- D. 20

Correct Answer: B

Detailed Solution: One-vs-All requires N classifiers for N classes.

Question 7

Which of the following statements about kernel functions are true?

1. Kernels allow SVMs to handle non-linear decision boundaries.
2. Kernels compute similarity between data points.
3. Kernels always reduce dimensionality.

- A. 1 only
- B. 1 and 2 only
- C. 2 and 3 only
- D. 1, 2 and 3

Correct Answer: B

Detailed Solution: Statements 1 and 2 are true.

Question 8

Suppose the sigmoid function output is $g(z)=0.6$. What is the value of the gradient of $g(z)$ with respect to z (i.e. $g'(z)$)?

- A. 0.12
- B. 0.24
- C. 0.36
- D. 0.48

Correct Answer: B

Detailed Solution: $g'(z) = g(z)*(1-g(z)) = 0.6 \times 0.4 = 0.24$.

Question 9

Which kernel is commonly used when the classes are not linearly separable and the shape of the boundary is unknown?

- A. Linear Kernel
- B. Polynomial Kernel (degree 1)
- C. Gaussian (RBF) Kernel
- D. No Kernel

Correct Answer: C

Detailed Solution: RBF kernels can model complex nonlinear boundaries.

Question 10

Which of the following statements are true?

1. Logistic Regression is a linear classifier.
2. SVM can be used for classification tasks.
3. Kernel methods can help classify non-linearly separable data.

- A. 1 only
- B. 2 and 3 only
- C. 1 and 2 only
- D. 1, 2 and 3

Correct Answer: D

Detailed Solution: All statements are true.
